
The Critical Listener

An Embedding-Based Music Recommender Built on Music Criticism

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Erdős Institute, Deep Learning Bootcamp, Summer 2026

Introduction

- Music industry recommendation algorithms by companies such as Spotify optimize for continuous engagement rather than music exploration
- Critics have wide-ranging musical knowledge, thus enabling the identification of unusual yet apt connections between musical works
- **Central Question:** Can semantic similarity between critic reviews rather than user listening behavior surface meaningful album connections that help curious listeners venture beyond their typical taste profile?



The Critical Listener

GOAL

Use natural language processing (NLP) to build a recommender system where albums are matched by the semantic approach reviewers use to describe them

NOVELTY

- Makes recommendations based on critical language rather than listening behavior
- More unexpected connections than industry recommenders
- Includes LLM-based “explainer” of album matches

STAKEHOLDERS

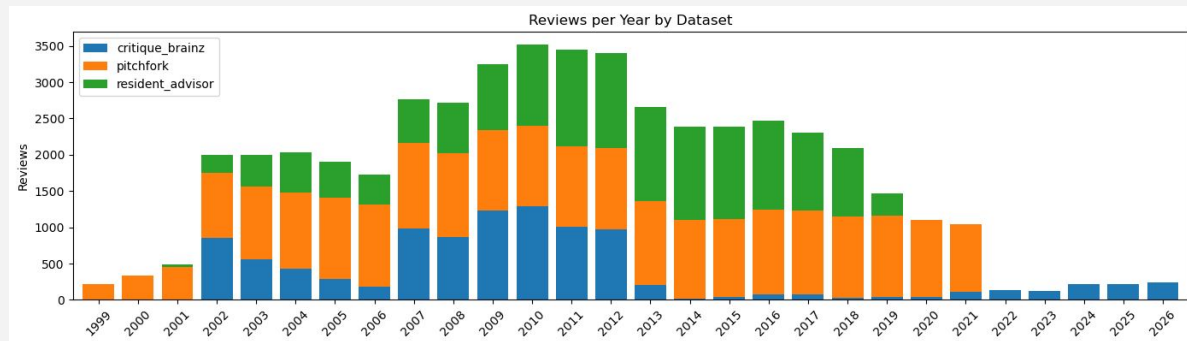
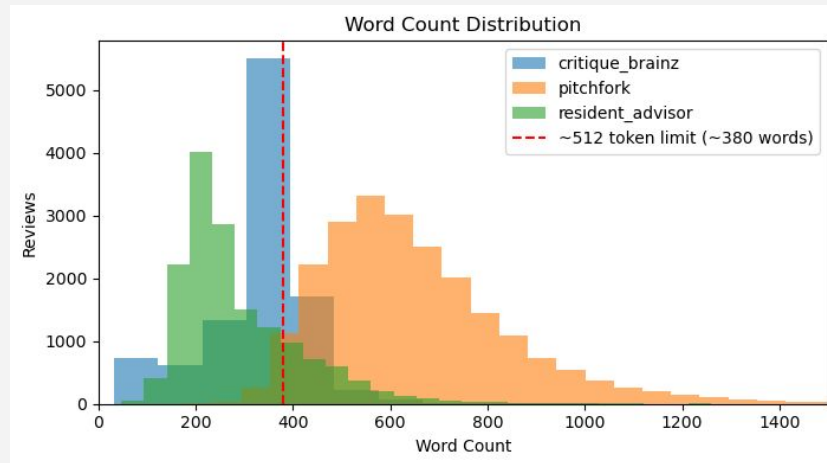
- Music lovers & musicians
- Journalists & music publications
- Music industry (streaming services, record companies)

Dataset

48,639 reviews of 44,717 unique albums

Sources

- Pitchfork (22,853 professional reviews, 1999-2021)
- Resident Advisor (15,560 professional reviews, 2001-19)
- CritiqueBrainz (10,226 professional and amateur reviews, 2002-26)



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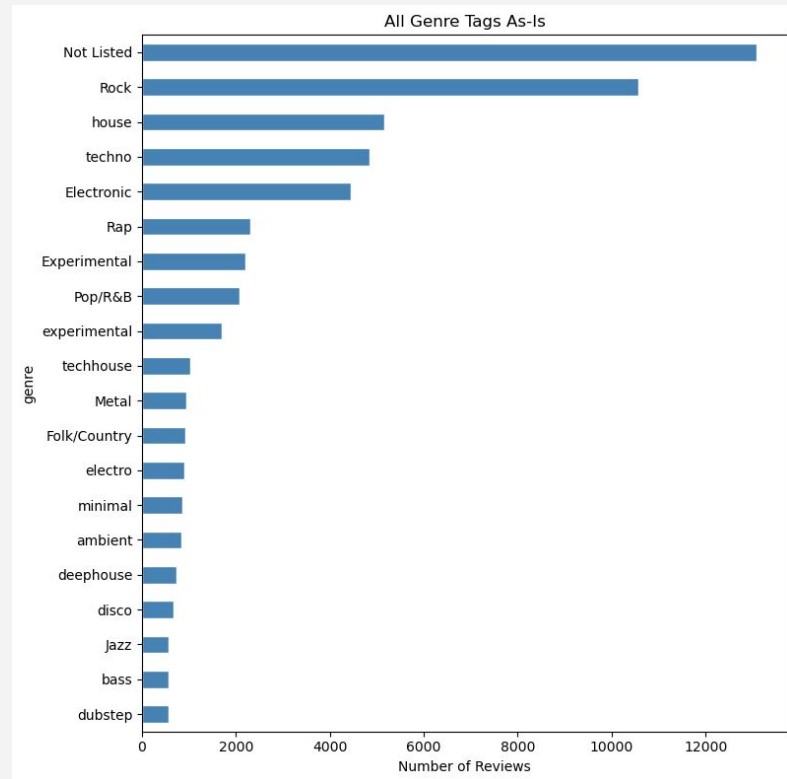
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Genres

- Highest frequency: Rock, Electronic
- Moderate frequency: Hip hop, Experimental, Pop, R&B
- Low frequency: Jazz, World, Folk/Country



Embedding-Based Recommender

EMBED REVIEWS

01

Turn each album review into a vector using Nomic embeddings.

AVERAGE TO ALBUM

02

Pool all review vectors for an album, average them, and normalize to one album embedding.

SCORE SIMILARITY

03

For a query album, compute cosine similarity against every other album in the catalog.

FILTER AND RANK

04

Drop the query and same-artist albums, then return the top k most similar albums.

Model Selection

Baseline: TF-IDF

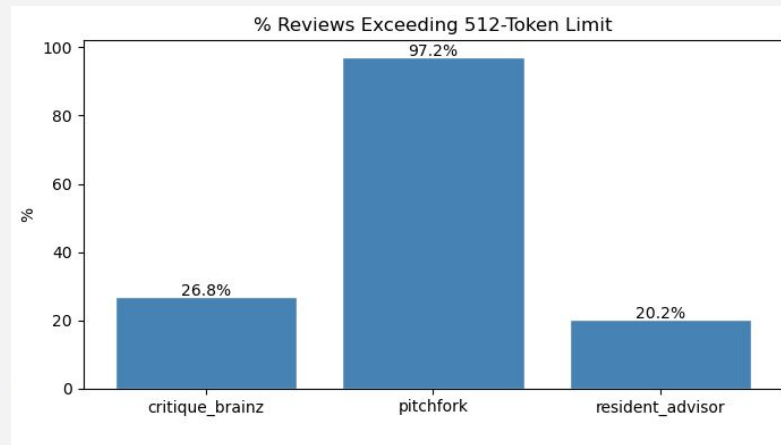
Compared models:

- MiniLM (512-token model)
- E5-large (512-token model designed to handle longer texts despite token limits)
- Nomic Embed Text 1.5 (8,192 token model, handles long text better) **(winner)**

Task formulation: retrieval (given a review of Album X by Reviewer A, can the model retrieve a review of Album X by Reviewer B as the top result?)

Preprocessing

- Replace artist names with “[MASKED]”
- Prepend artist & album name to review text



Model Comparison: Masked Input

Model	Context (tokens)	MRR	Recall@1	Recall@5	Recall@10
TF-IDF	—	0.881	81.8%	96.6%	98.2%
MiniLM	512	0.551	44.4%	67.4%	74.4%
E5-large	512	0.908	85.0%	97.8%	99.0%
Nomic	8,192	0.934	89.8%	98.2%	99.0%

The “Explainer”

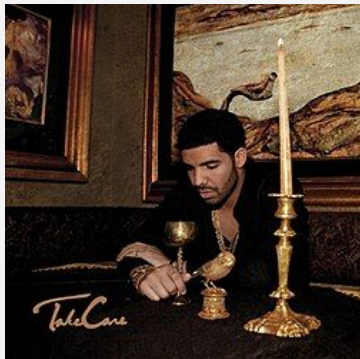
To demonstrate semantic similarity, we developed an LLM-based explainer that identifies shared themes and quotes both reviews side-by-side. Using a structured prompt via Claude Haiku, the explainer:

- Identifies up to 3 genuine shared qualities that must be grounded in verbatim quotes (no inference/hallucination)
- Demonstrates not only similarity between albums, but *why* a critic would connect the two
- Compensates for relative shortage of hard verification metrics
- Makes the case to the listener why they should listen to the new album

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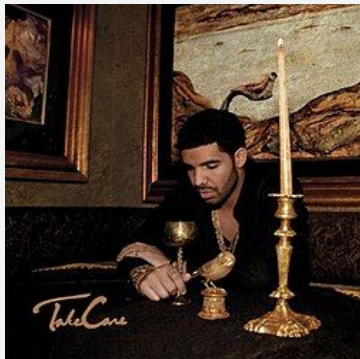
SAMPLE RECOMMENDATIONS:

Drake - *Take Care*
(2011)

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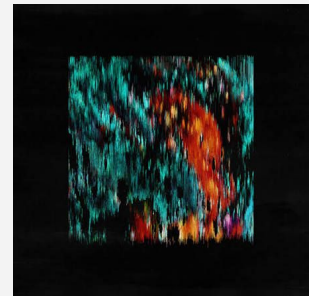
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#1: Stwo - D.T.S.N.T (2016)

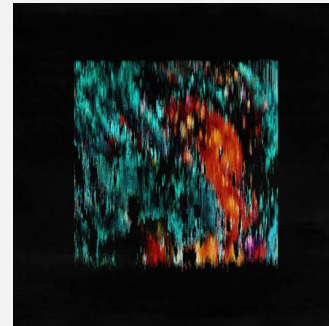
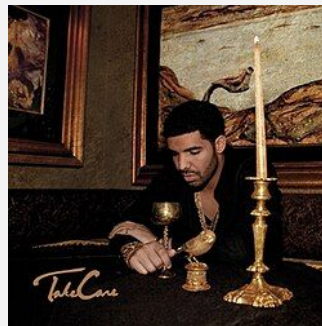
#2: Majid Jordan - Majid Jordan (2016)

#3: Partynextdoor - Partynextdoor (2013)



The “Explainer”

Similarities between Drake - *Take Care* & Stwo - *D.T.S.N.T.*



Production aesthetic of muted, atmospheric backdrops

"its muted backdrops - like listening to a record on your neighbour's stereo, through the dividing wall - were perfect for emoting over." (*Take Care*, CritiqueBrainz)

"The drums are programmed and flat, the mix is smoky and these beautiful voices float atop it all, creating a drugged-out haze" (*D.T.S.N.T.*, Pitchfork)

Collaboration with skilled producers to elevate artistic vision

"...by surrounding himself with the right producers and guests (and presumably the right management), he's elevated himself to remarkable heights" (*Take Care*, CritiqueBrainz)

"STWO...is worth seeking out if, like Drake, you're a vocalist looking to flex your abilities. On his new EP, *D.T.S.N.T.*, he makes room for a globally sourced array of collaborators..." (*D.T.S.N.T.*, Pitchfork)

Noah "40" Shebib's production influence

"Calling predominantly on the production skills of Noah '40' Shebib..." (*Take Care*, CritiqueBrainz)

"STWO's collaborators thrive over the kinds of beats that producers like Noah '40' Shebib, Clams Casino, and Shlohmo mastered long ago" (*D.T.S.N.T.*, Pitchfork)

Benchmark

Last.fm Album Recommender

The Last.fm logo, consisting of the text "last.fm" in white lowercase letters on a red rectangular background.

Last.fm is a music tracking service that can recommend songs based on listening habits of millions of users.

Given it only works at the track-level, we constructed an album-level recommender by aggregating track-pair scores into a album-to-album recommendations, providing a realistic, industry-style baseline for comparing against our recommender.

Benchmark

We generated the top-5 recommendations for 11,000 random albums in our catalog, resulting in 45,000 recommendations slots.

We compute 30+ metrics to evaluate recommendation quality, analyzing data at album, artist, and user tags-levels in several thematic categories.

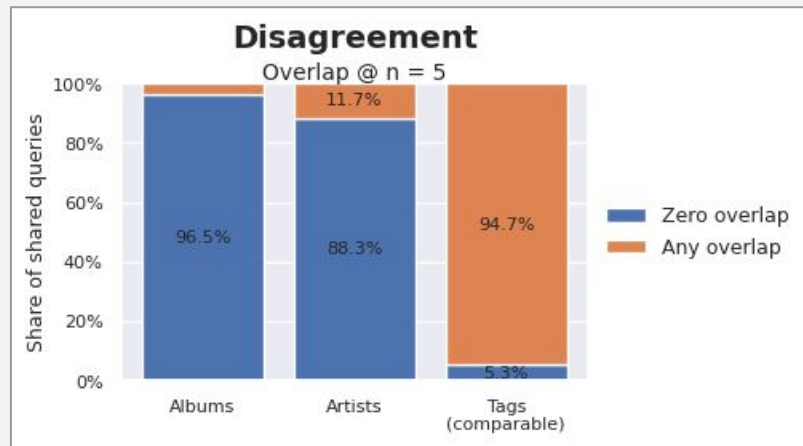
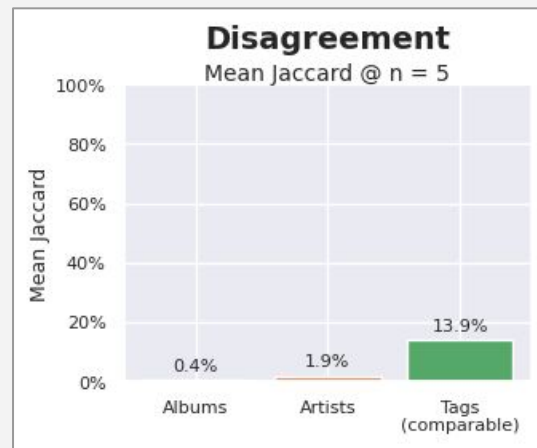
Benchmark

Last.fm Album Recommender

Theme	Ct	Metrics
Agreement	4	precision, recall, hit_rate, mean_intersection
Disagreement	3	mean_jaccard, zero_overlap_share, any_overlap_share
Diversity	4	mean_unique, mean_diversity_ratio, all_distinct_share, single_target_share
Novelty	6	mean_reach, singleton_share, mean_self_information, median_self_information, mean_novelty, effective_catalog
Popularity	8	Reviews: mean_reviews_recs, mean_reviews_catalog, mean_reviews_ratio, recs_3plus_share Listeners: mean_rec_listeners, mean_query_listeners, mean_rec_query_ratio, rec_more_popular_share
Reciprocity	1	reciprocal_rate
Repetition	5	hub_share, hub_slot_share, unique_share, gini, top1pct_concentration
Serendipity	2	serendipitous_share, serendipitous_listener_share

Disagreement

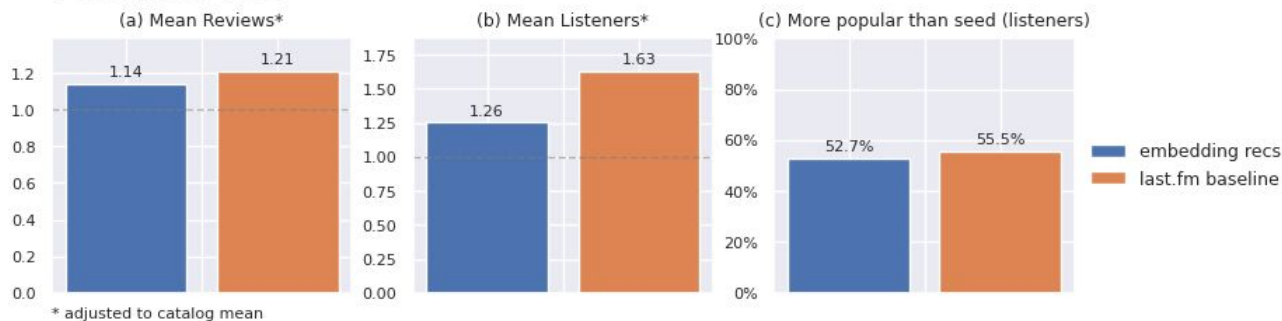
- Low **Jaccard similarity** shows that our recommender and Last.fm's largely diverge.
- High **share of any overlap** (or **Hit Rate@5**) within tags indicate divergence isn't random. Recommenders usually agree on at least one shared item.
- Together, this suggests our recommender captures the same baseline relevance as Last.fm, then diverges into the long tail, producing more diverse recommendations without sacrificing accuracy.



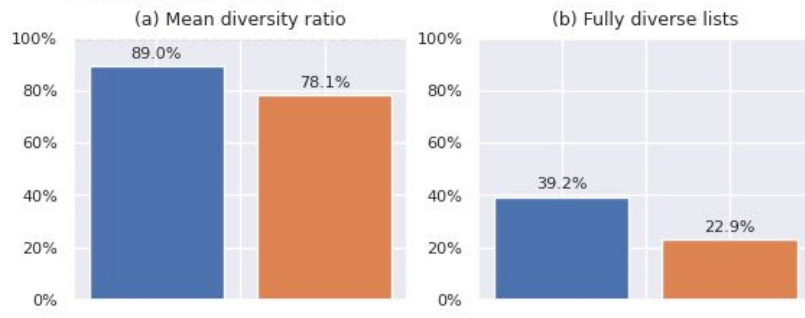
Beyond Accuracy

A comprehensive evaluation is available at [our GitHub repo](#).

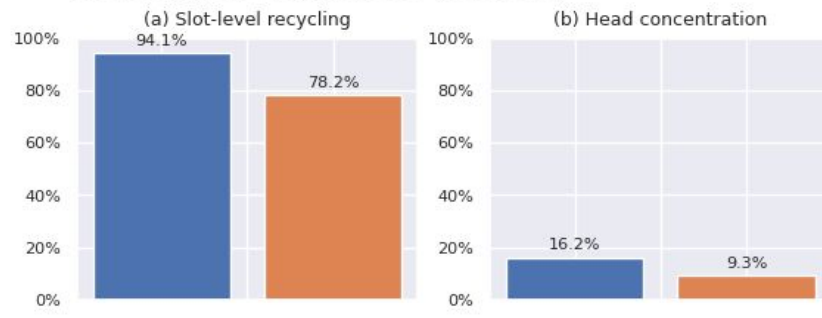
Popularity bias



Diversity (tags)



Repetition & hubness (artists)



Comparison

Top 3 Recommendations

Drake - Take Care (2011)

Embedding Model	Last.fm Recommender
Stwo - D.T.S.N.T French electronic/chillwave producer known for hazy, atmospheric instrumentals	J. Cole - Cole World: The Sideline Story Mainstream conscious hip hop debut from a chart-topping rapper
Majid Jordan - Majid Jordan Canadian R&B/synth-pop duo blending dreamy production with dance-pop textures	Drake & 21 Savage - Her Loss Collaborative mainstream rap album co-starring Drake himself
Partynextdoor - Partynextdoor Moody alt-R&B project built on minimalist, late-night production	Kanye West - Yeezus Top-charting hip hop album from an A-list rapper

Conclusion

- Our NLP-based recommender surfaces a wider range of music and relies less on popularity than industry-standard tools
- Strengths
 - Less popularity bias
 - More diverse recommendations than industry recommenders
 - Novel explainer function justifies selection with quotes from music critics
- Weaknesses
 - Because our model is based on *semantic* similarity rather than verified user listening patterns, we had to use indirect methods of comparison and validation
 - Review text limited to Pitchfork, Resident Advisor, and CritiqueBrainz
- Future work
 - Incorporate additional review datasets (e.g., publications such as The Wire, Consequence, Spin, Rolling Stone)
 - Explore additional NLP models
 - Incorporate additional guardrails that prevent superficial similarities and same-artist pairings.